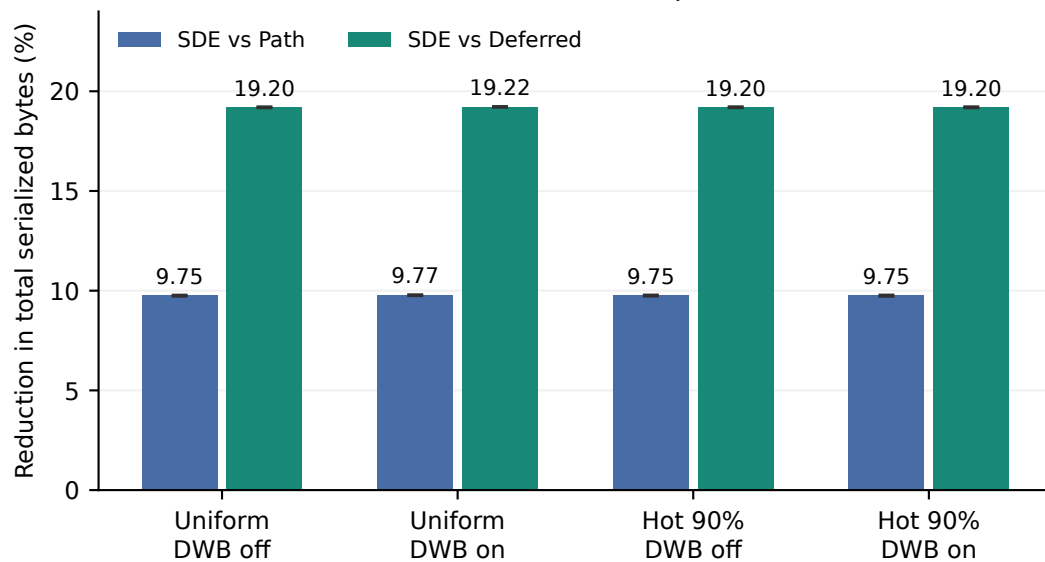
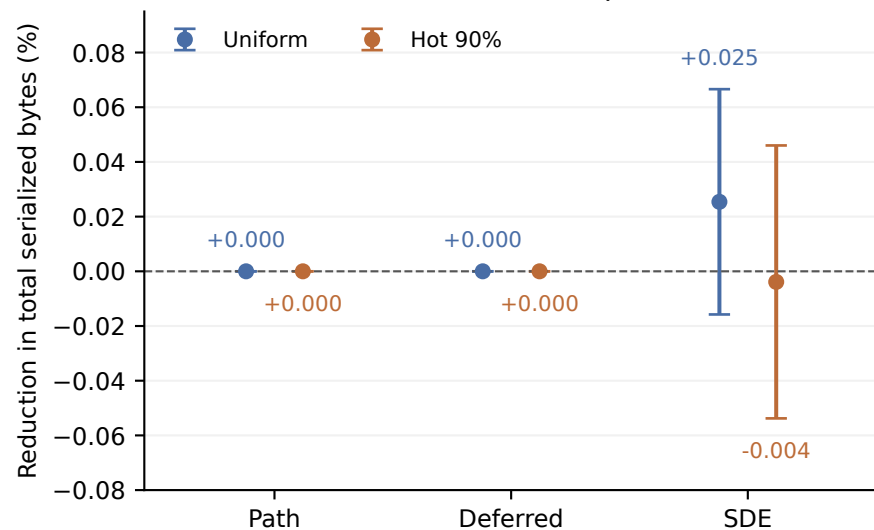


IR integrated pipeline: complete-period window

(a) Selective backend replacement



(b) DWB off to on: fixed public slots



N = 4,096; B = 64 B; depth profile; raw recursive maps + LLC + fixed-slot DWB.
1,536 warmup + 6,144 measured requests; 49,152 measured public slots; five paired runs; 95% t intervals.
Four Deferred/SDE service cycles; Path uses the same window. Initialization/alignment are billed separately.
Finite window, not a steady-state claim. Dirty LLC entries may remain. IR-Stash is not included.